Quantum Superposition Decay for Quantum Complexity Classes

A Classical–Quantum Bridge via the Read-Write-Execute Memory Model

Nnamdi Okpala

OBINexus Computing

okpalan@protonmail.com

April 9, 2026

Abstract

We introduce *Superposition Decay* (SD) as a formal measurement window that governs the collapse of a quantum superposition state on a classical computing substrate. Building on standard quantum postulates and the wave-function collapse principle, we show that any classical algorithm bounded by linear time complexity—deterministic or non-deterministic—can be evaluated in parallel via superposition decay on a quantum system. We further demonstrate that all problems belonging to a system of linear equations can be expressed as a probabilistic directed acyclic graph (DAG) on such a system. A concrete Read-Write-Execute (RWX) memory model formalises the decay window in operational terms: four write operations and two read operations reduce atomically to a single coherent read. We map these results onto the standard quantum complexity hierarchy ($\mathsf{P} \subseteq \mathsf{BPP} \subseteq \mathsf{BQP} \subseteq \mathsf{PP} \subseteq \mathsf{PSPACE}$) and characterise which promise-problem classes are tractable under superposition decay. The framework is then extended in two directions: (i) **Rift**, a quantum-classical grammar matcher whose parser exploits grammar superposition to evaluate all production rules in parallel, collapsing to a unique parse tree at a grammar decay event; and (ii) the **Quantum Burst Protocol (QBP)**, a communication protocol in which all candidate network paths are held in superposition during handshake and selected by a single measurement event. Both components are implemented within the MMUKO OS / OBINexus toolchain via the pipeline `riftlang.exe` → `.so.a` → `rift.exe` → `gosilang`, orchestrated by `nlink` and `polybuild`.

Contents

1 Introduction **2**

2 Preliminaries **3**
2.1 Quantum Postulates . . . 3
2.2 Classical Turing Machine and Complexity . . . 3
2.3 Quantum Circuits . . . 3
2.4 Probabilistic Directed Acyclic Graphs . . . 3

3 Superposition Decay **4**
3.1 Informal Motivation . . . 4
3.2 Formal Definition . . . 4
3.3 Constructive and Destructive Superposition . . . 4

4 The Read-Write-Execute (RWX) Memory Model **5**
4.1 Motivation . . . 5
4.2 Formal Model . . . 5
4.3 RWX Permission Table . . . 6

5 Complexity Classes Under Superposition Decay **6**
5.1 Promise Problem Formulation . . . 6
5.2 Complexity Class Definitions . . . 6
5.3 Tractability Under SD . . . 6

6 The Main Parallelism Theorem **8**
6.1 Statement . . . 8
6.2 Linear Systems as Parallel DAGs . . . 8

7 Rift: A Quantum-Classical Grammar Matcher **9**
7.1 Overview . . . 9
7.2 Wave Postulates as Triangular Matching Postulates . . . 9
7.3 Parallel Grammar Evaluation . . . 9
7.4 Stabilisation Principle . . . 10

8 Quantum Burst Protocol (QBP) **10**
8.1 Motivation . . . 10
8.2 Formal Definition . . . 10
8.3 QBP Handshake as Superposition Decay . . . 11
8.4 Spectral Graph Representation . . . 11

9 MMUKO OS and the OBINexus Toolchain **12**
9.1 MMUKO OS . . . 12
9.2 Toolchain Stack . . . 12

10 Conclusion and Future Work **13**
10.1 Future Work . . . 13

1 Introduction

The relationship between quantum and classical computation is most precisely characterised through the lens of promise-problem complexity classes. While quantum speedup for specific problem families (integer factorisation, unstructured search) is well established, a general framework that maps the *measurement event*—the moment a quantum superposition collapses to a definite classical outcome—onto classical computational primitives has received less systematic attention.

This paper proposes such a framework under the name **Superposition Decay** (SD). The central intuition is that the collapse of a superposition is not a passive observation but an active computational window: a bounded interval of control during which a system may coherently commit or roll back parallel computations, analogous to a transactional memory operation.

We formalise five related contributions:

C1. The SD **window.** A formal definition of the measurement window as a triple (resolved / rejected / pending), bounding acceptance and rejection by probability thresholds $a(n)$ and $b(n)$, with an intermediate *pending* region supporting deferred evaluation.

C2. The RWX memory model. A Read-Write-Execute abstraction that maps quantum measurement to classical memory transactions: n writes followed by k reads collapse to a single coherent read event when the last read is the terminal operation.

C3. Complexity-class tractability under SD**.** A characterisation showing that BQP promise problems are exactly the class tractable under a single superposition decay event; problems in PP and PSPACE require multiple decay events or additional measurement principles.

C4. Rift: quantum-classical grammar matching. A formalisation of the Rift compiler's parser as a quantum automaton, with grammar production rules as superposed branches and parse commitment as a grammar decay event governed by the stabilisation principle.

C5. Quantum Burst Protocol (QBP). A communication protocol in which all candidate network paths are held in superposition during connection negotiation and selected by a single measurement (spectral decay), achieving optimal routing in $O(1)$ measurement time.

The paper is organised as follows. Section 2 reviews necessary background in quantum mechanics and complexity theory. Section 3 defines superposition decay formally. Section 4 presents the RWX memory model. Section 5 maps results onto the complexity hierarchy. Section 6 states and proves the main parallelism theorem. Section 7 formalises Rift as a quantum-classical grammar matcher. Section 8 introduces QBP. Section 9 describes the MMUKO OS and OBINexus toolchain. Section 10 concludes.

2 Preliminaries

2.1 Quantum Postulates

We work in the standard Hilbert-space formulation of quantum mechanics. A quantum state is represented by a unit vector $|\psi\rangle \in \mathbb{C}^{2^n}$ for an n-qubit system.

Axiom 2.1 (State Postulate)**.** The complete state of a quantum system at time t is described by a normalised vector $|\psi(t)\rangle \in \mathcal{H}$, the system's Hilbert space, satisfying $\langle \psi(t) \mid \psi(t)\rangle = 1$.

Axiom 2.2 (Superposition Postulate)**.** If $|\psi_1\rangle$ and $|\psi_2\rangle$ are valid states of a quantum system, then any linear combination

$$|\psi\rangle = c_1 |\psi_1\rangle + c_2 |\psi_2\rangle , \quad c_1, c_2 \in \mathbb{C}, \quad |c_1|^2 + |c_2|^2 = 1,$$

is also a valid state. This follows from the linearity and homogeneity of the Schrödinger equation.

Axiom 2.3 (Wave-Function Collapse / Measurement Postulate)**.** Upon measurement of an observable $\hat{O}$ with eigenvalues $\{\lambda_i\}$ and eigenstates $\{|e_i\rangle\}$, the system transitions from $|\psi\rangle = \sum_i c_i |e_i\rangle$ to eigenstate $|e_j\rangle$ with probability $|c_j|^2$. The post-measurement state is $|e_j\rangle$.

Axiom 2.4 (Evolution Postulate)**.** Between measurements, a closed quantum system evolves unitarily: $|\psi(t)\rangle = U(t) |\psi(0)\rangle$, where $U(t)$ is a unitary operator.

2.2 Classical Turing Machine and Complexity

A *Turing machine* (TM) is a tuple $M = (Q, \Sigma, \Gamma, \delta, q_0, q_{\text{acc}}, q_{\text{rej}})$ with the standard interpretation. Time complexity $T(n)$ denotes the maximum number of steps on inputs of length n. We use the standard complexity classes P, NP, BPP, PP, PSPACE, and BQP; their promise-problem definitions are recalled in Section 5.

2.3 Quantum Circuits

A *quantum circuit* Q_n on n qubits is a sequence of gates drawn from a universal gate set (e.g., $\{H, \text{CNOT}, T\}$) that accepts an n-qubit input and produces a 1-qubit output. A *uniform circuit family* $\mathcal{Q} = \{Q_n : n \in \mathbb{N}\}$ is one where the description of Q_n is computable in polynomial time from n.

2.4 Probabilistic Directed Acyclic Graphs

Definition 2.5 (Probabilistic DAG)**.** A *probabilistic directed acyclic graph* $G = (V, E, p)$ is a DAG where each edge $(u, v) \in E$ carries a probability weight $p(u, v) \in [0, 1]$ such that for every vertex u, $\sum_{v:(u,v)\in E} p(u, v) \leq 1$. Each path from a source to a sink represents a computation branch; the probability of a path is the product of its edge weights.

3 Superposition Decay

3.1 Informal Motivation

When a quantum system in superposition is measured, the resulting classical outcome is not predetermined—it is drawn from a probability distribution defined by the amplitudes. We argue that this *measurement event* is best understood not as an instantaneous point but as a *window*: a bounded region of the computation during which the system retains coherence and the outcomes of parallel branches may still be combined or discarded.

Once the window closes—the superposition *decays*—a single classical outcome is committed. The SD framework captures this window operationally.

3.2 Formal Definition

Definition 3.1 (Quantum Connection Graph)**.** A *quantum connection* is a graph $G = (E, N)$ where N is the set of n-qubit nodes (quantum registers) and $E \subseteq N \times N$ is the set of entanglement edges. A path from node A to node B in G represents a sequence of entangled operations connecting two computational loci.

Definition 3.2 (Superposition State Vector)**.** For an n-qubit system, the superposition state is

$$|\Psi\rangle = \sum_{x \in \{0,1\}^n} \alpha_x |x\rangle, \quad \alpha_x \in \mathbb{C}, \quad \sum_x |\alpha_x|^2 = 1.$$

The probability of observing basis state $|x\rangle$ is $p(x) = |\alpha_x|^2$.

Definition 3.3 (Superposition Decay Window)**.** Let $\mathcal{Q} = \{Q_n\}$ be a uniform quantum circuit family and let $a, b : \mathbb{N} \to [0, 1]$ be functions with $a(n) > b(n)$ for all n. The *superposition decay window* $\mathrm{SD}(a, b)$ is the triple

$$\mathrm{SD}(a, b) \;=\; \big(\text{resolved, rejected, pending}\big),$$

where for an input x of length n:

- x is **resolved** (accepted) if $\Pr[Q_{|x|}(x) = 1] \geq a(|x|)$,
- x is **rejected** if $\Pr[Q_{|x|}(x) = 1] \leq b(|x|)$,
- x is **pending** if $b(|x|) < \Pr[Q_{|x|}(x) = 1] < a(|x|)$.

The *decay event* is the transition of a state from *pending* to either *resolved* or *rejected* upon measurement.

Remark 3.4. The three-valued output (resolved / rejected / pending) generalises the standard binary outcome of a classical TM. The *pending* state corresponds precisely to the undecided region of a promise problem: the system is contractually guaranteed not to receive inputs from this region in the promise-problem formulation, but the SD framework makes the region explicit as a computational state.

3.3 Constructive and Destructive Superposition

The superposition principle admits both *constructive* interference (amplitudes add, increasing probability of a desired outcome) and *destructive* interference (amplitudes can-

cel, suppressing undesired outcomes). We formalise this in terms of the wave function as follows.

Definition 3.5 (Constructive / Destructive Interference)**.** Given two states $|\psi_1\rangle = \sum_x \alpha_x |x\rangle$ and $|\psi_2\rangle = \sum_x \beta_x |x\rangle$, their superposition $|\psi\rangle = c_1 |\psi_1\rangle + c_2 |\psi_2\rangle$ exhibits:

- **Constructive interference** at x if $|c_1\alpha_x + c_2\beta_x| > \max(|c_1\alpha_x|, |c_2\beta_x|)$.
- **Destructive interference** at x if $|c_1\alpha_x + c_2\beta_x| < \min(|c_1\alpha_x|, |c_2\beta_x|)$.

In the SD framework, a quantum algorithm exploits constructive interference to amplify the amplitude of the correct solution branch before the decay window closes, and destructive interference to suppress incorrect branches.

4 The Read-Write-Execute (RWX) Memory Model

4.1 Motivation

Classical memory access is characterised by read (R), write (W), and execute (X) permissions, typically represented as a triple (r, w, x) per memory region. We propose a quantum analogue in which the *sequence* of reads and writes within a superposition decay window determines the atomicity and coherence of the resulting classical observation.

4.2 Formal Model

Definition 4.1 (RWX Operation Sequence)**.** An *RWX sequence* is a finite ordered list of operations $\sigma = (o_1, o_2, \ldots, o_k)$ where each $o_i \in \{R, W, X\}$. The sequence is *terminal-read* if $o_k = R$ (the final operation is a read).

Definition 4.2 (Atomic Decay Unit)**.** A terminal-read RWX sequence of w writes followed by r reads,

$$\sigma = (\underbrace{W, W, \ldots, W}_{w},\ \underbrace{R, R, \ldots, R}_{r-1},\ R),$$

constitutes a single *atomic decay unit* (ADU) equivalent to one coherent read operation. Formally:

$$w \cdot W + r \cdot R \xrightarrow{\text{SD}} 1 \cdot R_{\text{coherent}}.$$

Example 4.3 (4-Write 2-Read ADU)**.** The concrete instance stated in the source document is:

$$4W + 2R \xrightarrow{\text{SD}} 1R.$$

This means: four parallel write operations (corresponding to four parallel computation branches in superposition) and two intermediate read probes collapse—upon superposition decay—into a single coherent classical read outcome. The dual-state nature of the system (write = superposed, read = collapsed) ensures that the terminal read is the only externally observable event.

Proposition 4.4 (Dual-State Equivalence)**.** *In a quantum system respecting the* SD *window, the state before the terminal read is a superposition of all write branches. The*

terminal read collapses this superposition to a single classical value. Therefore, the external observer sees exactly one read operation regardless of the number of internal write branches.

Proof. By the Measurement Postulate (Axiom 3), measurement of the system in state $|\Psi\rangle = \sum_x \alpha_x |x\rangle$ produces outcome x with probability $|\alpha_x|^2$ and collapses the state to $|x\rangle$. Each write branch corresponds to a basis state $|x\rangle$. Since the terminal read is the sole measurement, all prior writes are unobserved intermediate computations within the unitary evolution $U(t)$. By the Evolution Postulate (Axiom 4), $U(t)$ is reversible and does not produce classical outputs; only the terminal measurement does. Hence the external observable record contains exactly one read event. □ □

4.3 RWX Permission Table

The classical RWX permission system maps directly onto the decay window:

Permission	**Classical Meaning**	**Quantum** / SD **Analogue**
R (Read)	Load value from memory	Terminal measurement; collapses superposition
W (Write)	Store value to memory	Unitary gate application; maintains superposition
X (Execute)	Run code in memory region	Quantum circuit evaluation within decay window

5 Complexity Classes Under Superposition Decay

5.1 Promise Problem Formulation

A *promise problem* is a pair $(A_{\text{yes}}, A_{\text{no}})$ of disjoint languages. An algorithm *solves* the promise problem if it accepts every $x \in A_{\text{yes}}$ and rejects every $x \in A_{\text{no}}$, with no constraint on inputs outside $A_{\text{yes}} \cup A_{\text{no}}$.

5.2 Complexity Class Definitions

5.3 Tractability Under SD

Theorem 5.1 (SD-Tractability of BQP)**.** *A promise problem (A_{yes}, A_{no}) is in* BQP *if and only if it is solvable by a uniform quantum circuit family within a single superposition decay window* $\mathrm{SD}(a, b)$ *with $a(n) - b(n) \geq \frac{1}{3}$ for all n.*

Proof. ($\Rightarrow$) By definition of BQP, there exists a uniform circuit family $\{Q_n\}$ with the stated acceptance probabilities. The circuit Q_n operates entirely within a unitary evolution (all gates are write operations in RWX terms) and terminates with a single measurement (the terminal read). This constitutes a single ADU as in Definition 4.2, hence falls within one SD window.

($\Leftarrow$) Suppose a single $\mathrm{SD}(a, b)$ window with $a(n) - b(n) \geq \frac{1}{3}$ solves the promise problem. The circuit within the window is unitary (by the Evolution Postulate) and polynomial-time (by the polynomial-size constraint on the circuit family). This is exactly the definition of a BQP circuit. □ □

Class	Criteria
P	Promise problems for which a polynomial-time *deterministic* TM accepts all strings in A_{yes} and rejects all strings in A_{no}.
BPP	Promise problems for which a polynomial-time *probabilistic* TM accepts every string in A_{yes} with probability $\geq \frac{2}{3}$ and every string in A_{no} with probability $\leq \frac{1}{3}$.
BQP	Promise problems such that for functions $a, b : \mathbb{N} \to [0, 1]$ with $a(n) \geq \frac{2}{3}$ and $b(n) \leq \frac{1}{3}$, there exists a uniform quantum circuit family $\mathcal{Q} = \{Q_n : n \in \mathbb{N}\}$ where Q_n accepts n qubits and gives a 1-qubit output, such that every $x \in A_{\text{yes}}$ is accepted with probability $\geq a(\lvert x\rvert)$ and every $x \in A_{\text{no}}$ is accepted with probability $\leq b(\lvert x\rvert)$.
PP	Promise problems for which a polynomial-time probabilistic TM accepts every string in A_{yes} with probability $> \frac{1}{2}$ and every string in A_{no} with probability $\leq \frac{1}{2}$.
PSPACE	Promise problems for which a *deterministic* TM running in polynomial *space* accepts all A_{yes} and rejects all A_{no}.

Table 1: Complexity class definitions in the promise-problem framework.

Corollary 5.2. P $\subseteq$ BPP $\subseteq$ BQP *under the* SD *framework, since deterministic and bounded-error probabilistic computations can be simulated by a single decay window with all amplitude concentrated on one branch.*

Theorem 5.3 (Multi-Decay Requirement for PP)**.** *There exist promise problems in* PP $\setminus$ BQP *that require more than one superposition decay event to resolve under* SD*.*

Proof Sketch. PP allows acceptance probabilities arbitrarily close to $\frac{1}{2}$, whereas BQP requires a gap of at least $\frac{1}{3}$. A problem instance where $\Pr[\text{accept}] = \frac{1}{2} + \epsilon$ for arbitrarily small $\epsilon > 0$ cannot be distinguished from a reject instance (with $\Pr = \frac{1}{2} - \epsilon$) by a single decay event with fixed thresholds a, b satisfying $a - b \geq \frac{1}{3}$. Multiple sequential decay events (with amplitude amplification between events) are required to drive the probability above the BQP threshold. □ □

Remark 5.4. PSPACE problems, being characterised by polynomial-space deterministic computation, lie outside the reach of a constant number of superposition decay events unless BQP = PSPACE, which is not currently believed. They require fundamentally different principles—either unbounded decay depth or oracle access—within the SD framework.

6 The Main Parallelism Theorem

6.1 Statement

Theorem 6.1 (Classical-to-Quantum Parallelism via SD)**.** *Let $\mathcal{A}$ be a classical algorithm with time complexity $O(f(n))$ where f is a linear function ($f(n) = cn$ for some constant $c > 0$), and let $\mathcal{A}$ be deterministic or non-deterministic. Then there exists a quantum circuit family $\mathcal{Q}_{\mathcal{A}} = \{Q_n\}$ such that:*

1. *Q_n simulates all execution branches of $\mathcal{A}$ on inputs of length n in* parallel *within a single superposition decay window* $\mathrm{SD}(a, b)$.
2. *The circuit size of Q_n is polynomial in n.*
3. *The correct output is obtained upon the terminal measurement (the decay event) with probability $\geq \frac{2}{3}$.*

Proof. We construct Q_n as follows.

Step 1: Encode branches. The k non-deterministic branches of $\mathcal{A}$ on input x correspond to basis states $|b_1\rangle, \ldots, |b_k\rangle$ in a $\lceil \log_2 k \rceil$-qubit register. Since $\mathcal{A}$ runs in linear time $O(n)$, the number of branches is at most $k \leq 2^{cn}$ for some constant c; encoding requires $O(n)$ qubits.

Step 2: Superpose branches. Apply a Hadamard transform $H^{\otimes \lceil \log_2 k \rceil}$ to create the uniform superposition $|\Psi_0\rangle = \frac{1}{\sqrt{k}} \sum_{i=1}^{k} |b_i\rangle$.

Step 3: Parallel evaluation. By the linearity of unitary evolution, the circuit $U_{\mathcal{A}}$ that implements one step of $\mathcal{A}$ acts on all branches simultaneously: $U_{\mathcal{A}} |\Psi_0\rangle = \frac{1}{\sqrt{k}} \sum_i U_{\mathcal{A}} |b_i\rangle$. Since $\mathcal{A}$ runs in $O(n)$ steps, $U_{\mathcal{A}}$ consists of $O(n)$ layers, each implementable by a polynomial-size circuit.

Step 4: Amplitude amplification. Apply Grover-type amplitude amplification [1] to boost the amplitude of accepting branches. After $O(\sqrt{k})$ iterations, the probability of observing an accepting branch exceeds $\frac{2}{3}$.

Step 5: Terminal measurement. Measure the output qubit. This constitutes the single superposition decay event. By Theorem 5.1, the resulting problem is in BQP.

The total circuit size is $O(n \cdot \sqrt{k}) \leq O(n \cdot 2^{cn/2})$, which is polynomial for fixed $c < \frac{2 \log n}{n}$—i.e., for linear $f(n) = cn$ with small c. This covers all algorithms with sub-exponential branch counts, which includes all deterministic and polynomially-branching non-deterministic linear-time algorithms. □ □

6.2 Linear Systems as Parallel DAGs

Theorem 6.2 (Linear Systems as Probabilistic DAGs)**.** *Any system of m linear equations in n variables over $\mathbb{R}$ (or $\mathbb{C}$) with a unique solution can be expressed as a probabilistic DAG $G = (V, E, p)$ whose parallel evaluation under* SD *yields the solution in time $O(\log n)$ on a quantum system.*

Proof Sketch. The system $A\mathbf{x} = \mathbf{b}$ (with $A \in \mathbb{R}^{m \times n}$) has a unique solution $\mathbf{x}^* = A^{-1}\mathbf{b}$ (assuming A is invertible). The HHL algorithm [2] solves this system in $O(\log n \cdot \kappa^2)$ time

on a quantum computer, where κ is the condition number of A. We model each variable as a DAG node and each dependency as a directed edge. Gaussian elimination induces a topological order on the DAG; each elimination step is a parallel unitary operation on the corresponding qubit register. The terminal measurement on each variable register constitutes a separate SD event, and the results compose to give $\mathbf{x}^*$. □ □

7 Rift: A Quantum-Classical Grammar Matcher

7.1 Overview

Rift (formally: *RIFT — a Flexible Translator*) is the compiler front-end component of the OBINexus toolchain. Its theoretical grounding in the SD framework arises from the observation that grammar matching in a parser shares the structural properties of quantum superposition: multiple production-rule paths may be simultaneously *active* (pending) until sufficient lookahead disambiguates them. The decay event corresponds to the moment the parser commits to a single parse path.

7.2 Wave Postulates as Triangular Matching Postulates

The wave postulate (Axiom 3) asserts that a quantum system in superposition $|\Psi\rangle = c_1 |\psi_1\rangle + c_2 |\psi_2\rangle$ yields one of two outcomes upon measurement. We draw a direct structural analogy to a context-free grammar $G = (V, \Sigma, R, S)$ with two competing production rules $A \to \alpha \mid \beta$. In both cases:

- The system is in a *pending* superposition of both alternatives until a discriminating token (the "measurement") is encountered.
- The collapse to a single rule (the "decay") is the parser's commit to one production.
- Destructive interference corresponds to the parser eliminating impossible parse paths via lookahead or semantic constraints.

Definition 7.1 (Grammar Superposition State)**.** Let $G = (V, \Sigma, R, S)$ be a context-free grammar and let π be a partial parse of input $w = w_1 w_2 \cdots w_n$. The *grammar superposition state* at position i is the set

$$\Gamma_i \;=\; \left\{ (A \to \alpha \bullet \beta,\, j) \;\mid\; A \to \alpha\beta \in R,\; w_j \cdots w_i \Rightarrow^* \alpha \right\}$$

of all active Earley items. The state Γ_i is the grammar analogue of the superposition state vector: it encodes all simultaneously possible parse paths.

Definition 7.2 (Grammar Decay Event)**.** A *grammar decay event* at position i occurs when $|\Gamma_i| = 1$, i.e., all but one parse path has been eliminated by lookahead. This corresponds to the collapse of the grammar superposition state to a single production, exactly as measurement collapses $|\Psi\rangle$ to a single basis state.

7.3 Parallel Grammar Evaluation

Theorem 7.3 (Rift Parallel Parsing)**.** *Let G be an unambiguous context-free grammar and let w be an input string of length n. Rift evaluates all production rules of G in* parallel *by maintaining the full grammar superposition state Γ_i at each position, collapsing (decaying)*

only when the lookahead forces $|\Gamma_i| = 1$. The resulting parse is produced in $O(n^3)$ time in the worst case (Earley) and $O(n)$ for unambiguous grammars in Chomsky Normal Form, corresponding to a single-decay-event parse.

Proof Sketch. The Earley algorithm maintains all active items in Γ_i at each position. For an unambiguous grammar, the number of active items is bounded by a constant at each position after the initial scan, so the grammar superposition state decays to a singleton in $O(n)$ steps. This is structurally identical to a BQP computation: the circuit evaluates all branches in parallel and the terminal measurement (grammar decay) yields the unique correct parse with probability 1. □ □

7.4 Stabilisation Principle

The SD framework introduces a *stabilisation principle* as the binding mechanism between the quantum measurement window and its cognitive or computational substrate.

Definition 7.4 (Stabilisation Principle)**.** A system S obeys the *stabilisation principle* if, within the SD window, all intermediate states are reversible (unitary), and the terminal measurement commits the system to a stable classical output that cannot be further modified within the same decay window. Formally:

$$\forall\, t < t_{\text{decay}} : \; U(t) \text{ is unitary and reversible.} \qquad \text{At } t_{\text{decay}} : \text{ output is classical and irreversible.}$$

In the Rift context, this means: all grammar transitions before the decay event are reversible (backtracking is permitted); after the grammar decay event, the committed parse tree is final.

8 Quantum Burst Protocol (QBP)

8.1 Motivation

The *Quantum Burst Protocol* (QBP) extends the SD framework from computation to communication. Just as a quantum circuit collapses a superposition of computation branches to a single classical output, a QBP connection collapses a superposition of communication paths between two endpoints to a single established channel.

8.2 Formal Definition

Definition 8.1 (QBP Connection)**.** A *Quantum Burst Protocol connection* is a tuple

$$\text{QBP}(A, B) \;=\; \big(G,\; \text{SD}(a, b),\; \Pi\big),$$

where:

- $G = (E, N)$ is the quantum connection graph (Definition 3.1), with $A, B \in N$ the sender and receiver endpoints.
- $\text{SD}(a, b)$ is the superposition decay window governing the connection establishment handshake.

- $\Pi = \{\pi_1, \pi_2, \ldots, \pi_k\}$ is the set of candidate communication paths (edges in E) held in superposition during the handshake phase.

The connection is *established* (resolved) when the SD window closes and a single path $\pi^* \in \Pi$ is selected; it is *refused* (rejected) if no path survives the decay; and it is *negotiating* (pending) during the handshake phase.

8.3 QBP Handshake as Superposition Decay

Proposition 8.2 (QBP Three-Phase Handshake)**.** *A QBP connection establishment proceeds in three phases corresponding exactly to the three states of the* SD *window:*

1. ***Negotiating (pending):*** *Both A and B maintain a superposition of all feasible communication paths Π. No classical commitment is made.*
2. ***Resolved:*** *The* SD *window closes; the path π^* with maximal probability amplitude is selected. A and B are classically connected via π^*.*
3. ***Rejected:*** *The* SD *window closes with all path amplitudes below the threshold $b(n)$; no connection is established.*

Remark 8.3. The QBP handshake is a quantum analogue of the classical TCP three-way handshake (SYN / SYN-ACK / ACK), with the key difference that in QBP, *all candidate paths are simultaneously active* during the negotiating phase, and the path selection is determined by a single quantum measurement (the decay event) rather than sequential message exchange.

8.4 Spectral Graph Representation

The candidate path set Π is naturally represented as a *spectral graph*:

Definition 8.4 (Spectral Connection Graph)**.** The *spectral connection graph* of a QBP connection is the directed graph $\hat{G} = (V, E, \lambda)$ where $\lambda : E \to [0, 1]$ assigns each edge a probability amplitude (eigenvalue weight) such that $\sum_{e \in \Pi} \lambda(e)^2 = 1$. The selected path π^* is the eigenvector corresponding to the dominant eigenvalue after the decay measurement.

Example 8.5 (Internet Connection as QBP)**.** Consider connecting a client A to a server B over a network with k possible routing paths. In classical networking, these paths are evaluated sequentially (or via parallel probes with separate classical outcomes). Under QBP, the network interface holds all k paths in superposition:

$$|\Pi\rangle = \frac{1}{\sqrt{k}} \sum_{i=1}^{k} |\pi_i\rangle \,.$$

Quality-of-service metrics (latency, bandwidth, packet loss) act as the observable $\hat{O}$. Measurement of $\hat{O}$ collapses $|\Pi\rangle$ to the optimal path $|\pi^*\rangle$ in a single decay event, achieving optimal routing in $O(1)$ measurement time (assuming the superposition can be prepared in $O(\log k)$ time).

9 MMUKO OS and the OBINexus Toolchain

9.1 MMUKO OS

MMUKO OS (M for Mike, Q for Bella [Queen], K for Kilo, O for Oscar) is the operating system substrate for the OBINexus quantum-classical computing framework. The name encodes its philosophical foundation: *"the spirit of good and evil that connects nothing and everything, that binds all of us and all of them."* This duality—nothing and everything, good and evil—maps directly onto the binary superposition state $|0\rangle$ and $|1\rangle$ and their linear combinations.

MMUKO OS is a cloud-based Windows system that serves as the classical substrate on which quantum algorithms are emulated prior to the availability of dedicated quantum hardware. Its design is guided by three principles:

1. **Read-Write symmetry:** All memory operations respect the RWX model (Section 4), ensuring that write-heavy workloads collapse atomically to coherent read outcomes.
2. **Superposition-aware scheduling:** The OS scheduler treats parallel threads as superposed branches of a single computation, deferring classical commitment until the SD window closes.
3. **Decay-triggered synchronisation:** Inter-process communication is governed by QBP handshakes (Section 8), replacing traditional lock-based synchronisation with measurement-based collapse events.

9.2 Toolchain Stack

The OBINexus toolchain implements the SD framework across the full compilation pipeline:

Component	**Role in SD Framework**
`riftlang.exe`	Source language compiler; implements grammar superposition (Section 7) to parse Rift source files in parallel.
`.so.a`	Shared object / static archive; intermediate representation holding the superposed (pre-decay) compilation state.
`rift.exe`	Linked executable; represents the post-decay committed classical binary output.
`gosilang`	Target runtime language binding; maps SD primitives to hardware-level instructions.
`nlink`	Build linker / orchestrator; manages the RWX transaction across compilation units.
`polybuild`	Polyglot build system; coordinates multiple SD decay events across heterogeneous language targets.

The pipeline can be summarised as:

$$\texttt{riftlang.exe} \rightarrow \texttt{.so.a} \rightarrow \texttt{rift.exe} \rightarrow \texttt{gosilang} \xrightarrow{\texttt{nlink / polybuild}} \text{deployed binary.}$$

10 Conclusion and Future Work

We have formalised the concept of *Superposition Decay* as a measurement window bridging quantum and classical computation. The main results are:

- The SD window is a three-valued construct (resolved / rejected / pending) that generalises the binary output of a classical TM and aligns precisely with the promise-problem formulation of quantum complexity classes.
- The RWX memory model provides an operational interpretation of the decay window: n write operations and k read operations within the window collapse atomically to a single coherent read.
- BQP is exactly the class of promise problems solvable within a single SD event with a probability gap of $\frac{1}{3}$.
- Classical algorithms bounded by linear time complexity—deterministic or non-deterministic—can be simulated in parallel within a single SD window by a polynomial-size quantum circuit.
- Systems of linear equations can be expressed as probabilistic DAGs whose parallel evaluation under SD yields solutions in $O(\log n)$ quantum time.
- Rift's parser exploits grammar superposition to evaluate all production rules in parallel, collapsing to a unique parse tree at a grammar decay event governed by the stabilisation principle.
- The Quantum Burst Protocol (QBP) extends SD to networking: all candidate communication paths are held in superposition during handshake, with optimal path selection achieved by a single spectral decay measurement.
- The MMUKO OS / OBINexus toolchain implements all of the above across the full pipeline: `riftlang.exe` → `.so.a` → `rift.exe` → `gosilang`, orchestrated by `nlink` and `polybuild`.

10.1 Future Work

The following directions are planned within the OBINexus research programme:

1. **Hardware realisation.** Implementing the SD framework on near-term quantum hardware (superconducting qubits, trapped ions) via the MMUKO OS / `riftlang` compiler toolchain.
2. **Multi-decay protocols for PP and PSPACE.** Designing iterated SD protocols that amplify probability across multiple decay events to address problems beyond BQP.
3. **QBP networking stack.** Implementing QBP as a full networking protocol atop the MMUKO OS kernel, replacing TCP handshakes with spectral decay measurements for latency-optimal routing.

4. **Rift compiler validation.** Formal verification that the Rift grammar decay event produces parse trees equivalent to those of a classical Earley parser, with empirical benchmarks on `gosilang` target programs.
5. **Cognitive binding.** Exploring the stabilisation principle as a model for attention and commitment in cognitive architectures: the SD window as a formal model for the moment of decision in neural-symbolic systems.

References

[1] L. K. Grover, "A fast quantum mechanical algorithm for database search," *Proceedings of the 28th Annual ACM Symposium on Theory of Computing (STOC)*, pp. 212–219, 1996.

[2] A. W. Harrow, A. Hassidim, and S. Lloyd, "Quantum algorithm for linear systems of equations," *Physical Review Letters*, vol. 103, no. 15, p. 150502, 2009.

[3] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, 10th anniversary ed., Cambridge University Press, 2010.

[4] J. Watrous, "Quantum computational complexity," in *Encyclopedia of Complexity and Systems Science*, Springer, 2009.